Hybrid-Evidence Retrieval RAG Model for Hallucination Mitigation in AIGC Text Generation

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Abstract

Generative Artificial Intelligence (AIGC) large language models (LLMs) deliver powerful free-form text generation capabilities for intelligent writing, open-domain question-answering, and automated content creation. Nevertheless, the inherent hallucination phenomenon severely undermines the factual reliability of generated outputs, where LLMs produce linguistically fluent but factually fabricated statements, fake citations, and erroneous causal deductions. Conventional Retrieval-Augmented Generation (RAG) approaches still suffer from several critical drawbacks: semantic mismatch between retrieved evidence fragments and user queries, static equal weighting for all retrieved snippets, and noise interference introduced under long-context input scenarios. This paper proposes **HER-RAG**, a hybrid-evidence retrieval enhanced RAG framework dedicated to hallucination suppression for AIGC text generation. A two-layer retrieval pipeline combining sparse keyword-based retrieval and dense semantic vector retrieval is constructed to gather candidate reference evidence. An attention-driven dynamic evidence-weighting and filtering mechanism calculates factual relevance scores for each evidence snippet and discards low-confidence noisy fragments. Weighted high-quality evidence is injected as factual constraints into the large language model. Meanwhile, a factual-consistency auxiliary loss is introduced to penalize token outputs that conflict with reference materials during autoregressive generation. Hallucination suppression is collaboratively realized from both the retrieval side and the generation side. Evaluated on TruthfulQA and HotpotQA standard benchmarks, the proposed HER-RAG achieves a factual-consistency F1-score of 0.846 and hallucination-detection AUC of 0.881. Compared with the vanilla large-language-model baseline and conventional single-dense-retrieval RAG baseline, the hallucination occurrence ratio decreases by 22.7 % and 13.5 %, respectively. The proposed framework substantially reduces factual errors in AIGC-generated texts and improves

overall output trustworthiness, providing a feasible technical solution for high-reliability generative-AI content-creation applications.

Keywords: Generative Artificial Intelligence; AIGC; Large Language Model; Retrieval-Augmented Generation; Hallucination Mitigation; Factual Consistency; Evidence Retrieval

1. Introduction

Generative Artificial Intelligence (AIGC) built upon large language models has profoundly reshaped modern content-production workflows, supporting automatic composition, domain-oriented intelligent question-answering, and technical report drafting. Despite remarkable performance on multiple natural-language tasks, large language models suffer from the well-known hallucination defect. Hallucination refers to the phenomenon that models generate grammatically coherent texts containing fabricated facts, non-existent citations, incorrect numerical values, and false causal relationships. Such erroneous outputs impose severe misleading risks in high-stakes application domains including clinical healthcare, legal document assistance, and academic auxiliary writing [1].

Retrieval-Augmented Generation (RAG) has become one dominant technical paradigm for mitigating LLM hallucinations. The core idea of RAG is to retrieve factual reference documents from external structured or unstructured knowledge bases and feed the retrieved evidence together with user-input prompts into LLMs, so that the model generates responses grounded on real-world reference materials [2]. Nevertheless, existing conventional RAG frameworks still exhibit noticeable practical limitations. First, single dense-vector retrieval is prone to semantic drift and frequently returns text fragments that are lexically similar but factually irrelevant to the original user query. Second, nearly all evidence fragments retrieved are imported into the model prompt with identical static weights, without differentiating factual credibility and query-relevance; noisy or misleading evidence may introduce new factual mistakes into final outputs. Third, when processing long-context inputs containing massive evidence snippets, the large language model's internal attention mechanism tends to overlook critical reference information, and hallucination artifacts persist. Fourth, most existing RAG variants only modify the input prompt context and lack dedicated loss-function constraints targeting factual conflicts in the autoregressive generation phase.

Prior academic works have investigated multiple hallucination-mitigation technical routes. Lewis et al. [2] presented the foundational RAG architecture and validated that external knowledge retrieval can reduce hallucination frequency, yet their design adopts static fixed

evidence weights and lacks explicit noise-evidence filtering modules. Ji et al. [1] delivered a comprehensive survey covering diverse hallucination-detection methodologies, but most surveyed approaches belong to post-hoc detection schemes instead of in-generation source-side suppression. Several recent studies adopt hybrid sparse-dense retrieval pipelines; however, dynamic evidence-weighting and adaptive filtering mechanisms are seldom fully implemented [3]. Pure prompt-engineering-based hallucination-mitigation strategies can only achieve limited suppression effects and cannot fundamentally resolve model-inherent fabrication tendencies.

Research gap: End-to-end integrated frameworks that collaboratively suppress hallucinations across the full technical pipeline (evidence retrieval → evidence filtering-fusion → generation-phase factual loss constraint) remain insufficient. Few existing works simultaneously optimize retrieval-recall quality and impose explicit factual constraints at the token-generation stage for high-risk AIGC-text scenarios.

Major contributions of this paper are summarized as follows:

1. A two-layer hybrid-evidence retrieval module combining sparse BM25 keyword retrieval and dense semantic-vector retrieval is constructed to improve the recall rate of fact-relevant reference evidence.

2. An attention-driven dynamic evidence-weighting-and-filtering mechanism is designed. It computes semantic-factual relevance weights for each individual evidence fragment against user queries, filters out low-quality noisy evidence, and assigns dynamic importance weights for retained high-confidence snippets.

3. A factual-consistency auxiliary loss function is proposed and jointly optimized with vanilla LLM cross-entropy loss. This auxiliary loss imposes explicit penalties for tokens that contradict high-weight reference evidence during autoregressive token generation. Hallucination suppression is collaboratively accomplished on both the retrieval side and the generation side.

4. Comparative experiments and systematic ablation studies are conducted on TruthfulQA and HotpotQA public benchmarks to test hallucination-mitigation performance under various long-context input conditions.

The remainder of this paper is organized as follows. Section 2 reviews related theoretical background. Section 3 elaborates the HER-RAG overall architecture, two-layer hybrid retrieval workflow, dynamic evidence-fusion logic, and joint factual-consistency loss function. Section 4 describes experimental datasets, hardware-software environment, and evaluation metrics. Section 5 presents experimental results and detailed ablation analysis. Section 6 discusses model

limitations and promising future research directions. Section 7 concludes this paper. References are listed at the end of the manuscript.

2. Related Work

2.1 Large-language-model hallucination in AIGC

Large language models accomplish autoregressive token-by-token text generation after massive-scale text corpus pre-training. Hallucinations can be further categorized into two major classes [1]. Intrinsic hallucination means generated output contradicts information inside the given input context. Extrinsic hallucination denotes generated content that conflicts with objective real-world facts stored in external knowledge. Multiple root causes lead to hallucination artifacts: training-dataset noise, model knowledge cutoff dates, imperfect parameterized knowledge-memory capability, and internal context-understanding deviations. Existing hallucination-detection technologies include white-box and black-box detection paradigms. Detection-oriented methods can identify factual errors after generation finishes but cannot actively prevent erroneous token outputs during generation. This research focuses on source-side active suppression of hallucinated outputs.

2.2 Retrieval-Augmented Generation (RAG)

The standard RAG workflow contains a retrieval phase and a generation phase. During the retrieval phase, relevant document snippets are fetched from external knowledge bases. User-input queries concatenated with retrieved evidence fragments are fed as augmented prompts into large-language-model backbones, and the model generates answers grounded on retrieved reference materials [2]. Traditional RAG implementations mostly rely solely on single dense-vector retrieval, which leverages pre-trained embedding models to compute cosine-similarity matching between queries and corpus documents. This approach risks returning semantically-mismatched fragments. All retrieved snippets are indiscriminately appended into prompts, and noisy evidence may degrade final answer quality.

2.3 Sparse retrieval versus dense retrieval

- **Sparse retrieval (BM25)**: It operates based on keyword and term-frequency statistics. It demonstrates competitive performance for factual queries focusing on entities and proper nouns, but lacks deep semantic comprehension of paraphrased natural-language expressions.

- **Dense-vector retrieval**: Textual content is mapped into continuous vector space by pre-trained embedding models, and similarity is measured via cosine distance. It handles

semantically-paraphrased queries well but may return literally-similar yet fact-irrelevant text fragments.

These two retrieval paradigms exhibit complementary strengths; fusing them can effectively improve overall evidence-recall quality [3].

2.4 Factual consistency

Factual consistency evaluates the factual overlap degree between model-generated texts and ground-truth reference evidence. Core evaluation metrics include factual-consistency F1-score, hallucination-detection AUC, and hallucination-generation ratio. For high-stakes AIGC applications in healthcare and legal domains, factual-consistency metrics possess higher priority than conventional general-purpose text-generation metrics such as BLEU and ROUGE-L [4].

2.5 Practical pain-points of real-world AIGC text generation

1. Retrieved irrelevant or noisy evidence may conversely induce new hallucination artifacts instead of improving output quality.

2. Under long-context input conditions, massive mixed evidence fragments disperse LLM attention weights, and critical reference information tends to be ignored.

3. Most existing RAG implementations only modify input prompts, without designing dedicated loss-function constraints for factual mistakes occurring inside the generation procedure.

4. Complex multi-hop-reasoning queries pose substantial challenges for complete key-evidence recall when only a single-type retriever is deployed.

3. Methodology: HER-RAG Hybrid-Evidence Retrieval Hallucination-Suppression Model

The HER-RAG framework consists of four core functional modules: (1) two-layer hybrid-evidence retrieval module; (2) dynamic-evidence weighting-and-filtering module; (3) weighted-evidence prompt-construction module; (4) factual-consistency joint-loss generation module.

Overall workflow: User natural-language query is received → two-layer hybrid retrieval obtains sparse-retrieved and dense-retrieved candidate-evidence sets → evidence deduplication, attention-driven dynamic weighting calculation, filtering of low-weight noisy evidence → weighted-evidence-augmented prompt is constructed and fed into the large-language-model backbone → autoregressive generation is optimized by combining vanilla cross-entropy loss plus factual-consistency auxiliary loss → factual-enhanced AIGC-text answer

is output.

3.1 Two-layer hybrid-evidence-retrieval module

Two parallel retrieval branches are executed simultaneously:

1. Sparse BM25 retrieval: the BM25 branch returns the top-N_1 candidate evidence fragments from the external knowledge base using term- and entity-matching signals.

2. Dense-vector retrieval: a pre-trained embedding model encodes the user query and knowledge-base passages; the top-N_2 candidate evidence fragments are returned according to vector similarity.

AUTHOR ACTION REQUIRED: Insert the actual N_1 and N_2 values, embedding model, similarity metric, document chunk size/overlap, BM25 implementation, FAISS index type, and retrieval-corpus construction procedure used in the experiments.

Results originating from both retrieval branches are merged, and duplicate fragments are removed to obtain the combined candidate-evidence pool:

\begin{equation}
D={d_1,d_2,...,d_m}
\end{equation}

The complementary advantages of BM25 and dense-vector retrieval are exploited to enhance recall performance for entity-proper-noun queries as well as semantically-paraphrased natural-language queries.

3.2 Attention-driven dynamic-evidence weighting and filtering

For each candidate evidence fragment d_i, the query Q and evidence fragment are processed by the evidence-scoring module to obtain a relevance score w_i. Retained evidence is filtered and ranked using this score before prompt construction.

AUTHOR ACTION REQUIRED: Define AttentionScore(Q,d_i) precisely: specify the scoring architecture/model, input representation, normalization, training objective (if learned), score range, and threshold w_th. "AttentionScore" cannot remain an undefined black-box function in a reproducible journal manuscript.

3.3 Weighted-evidence prompt construction

The original user-input query Q together with filtered-and-weighted evidence collection are assembled to construct the final augmented input prompt. Each evidence snippet is annotated with its pre-computed relevance score, to guide the LLM to prioritize information from

high-weight reference fragments.

> Prompt template example:
>
> User Query: {Q}
>
> Reference Evidence (Relevance Score: w_i):
>
> Evidence 1: d_1 (Score: w_1)
>
> Evidence 2: d_2 (Score: w_2)
>
> Please strictly answer based only on the above-listed reference evidence. If no relevant information exists in the provided evidence, directly state "No relevant materials available". Fabricating facts is strictly prohibited.

3.4 Factual-consistency joint-loss function

The language-model training objective includes the standard autoregressive cross-entropy loss, L_CE. The proposed framework augments this objective with a factual-consistency term, giving L_total = L_CE + λL_FC.

AUTHOR ACTION REQUIRED: Provide the exact mathematical definition and implementation of L_FC. Explain how a generated token/sequence is determined to contradict evidence (e.g., the actual verifier/NLI/scoring mechanism), how gradients are obtained, the training data used for this objective, and the final λ value. This is central to the claimed novelty.

3.5 Inference-stage workflow (no loss-computation required during inference)

Pseudocode for HER-RAG inference

```
Plain Text
Input: User query Q, external knowledge-base KB, weight threshold w_th
Output: AIGC generated answer Answer
1: D_sparse ← BM25_retrieve(KB, Q, top-N1)
2: D_dense ← Dense_vector_retrieve(KB, Q, top-N2)
3: D ← D_sparse ∪ D_dense, perform evidence deduplication
4: for each d_i ∈ D do
5:     w_i ← AttentionScore(Q, d_i)
6: end for
7: Filter evidence fragments with w_i < w_th; sort remaining evidence descending by w_i
```

8: Construct weighted-evidence-augmented prompt

9: Answer ← LLM_generate(prompt)

10: return Answer

At inference time, no loss is computed. The retrieval, scoring/filtering, prompt-construction, and generation pipeline can operate without updating model weights. Compatibility with other language-model backbones should be described as a design objective unless it has been empirically validated on additional models.

4. Experiments

4.1 Datasets

1. **TruthfulQA**: A well-known hallucination-specialized evaluation dataset containing 817 carefully designed questions that tend to induce false factual outputs; each sample is paired with ground-truth correct answers and hallucinated reference examples [5].

2. **HotpotQA**: A multi-hop reasoning question-answering dataset that requires joint reasoning over multiple discrete evidence fragments, suitable for evaluating hallucination-suppression performance for complex multi-step-inference queries [6].

Base language-model backbone: Meta Llama 3 8B [7].
Hardware environment: RTX 4090 GPU and 32 GB system RAM.
Software stack: Python 3.10, LangChain, FAISS, and Hugging Face Transformers.
AUTHOR ACTION REQUIRED: Specify the exact Llama 3 checkpoint/version, tokenizer, embedding model, corpus/index construction, chunking strategy, N_1 /N_2 , evidence threshold, λ, optimizer, learning rate, epochs, batch size, decoding parameters, random seeds, and train/validation/test protocol. The current manuscript omits these values.

Evaluation metrics:

1. Factual-consistency F1-score

2. Hallucination-detection AUC, computed using the study's defined hallucination-labeling/scoring protocol.

AUTHOR ACTION REQUIRED: Define the hallucination ground truth and scoring procedure used to calculate AUC and hallucination ratio. State whether labels come from benchmark annotations, human raters, NLI/verifier models, LLM judging, or another method; include inter-rater agreement if human annotation was used.

3. Hallucination-generation ratio: percentage of test samples containing factual hallucination

outputs

4. ROUGE-L (general-purpose text-generation evaluation metric)

Baseline comparison methods:

- Baseline 1: Vanilla Llama-3-8B (direct generation without any RAG enhancement).
- Baseline 2: Conventional single-dense-retrieval RAG.
- Baseline 3: BM25-only sparse-retrieval RAG.
- Proposed: HER-RAG (two-layer hybrid retrieval + dynamic-evidence weighting + factual-consistency joint loss).

Ablation experimental groups:

① Remove two-layer hybrid retrieval; adopt single dense-vector retrieval only.

② Remove dynamic-evidence weighting-and-filtering; assign identical equal weights for all retrieved evidence fragments.

③ Remove factual-consistency auxiliary loss; utilize vanilla cross-entropy loss only.

④ Full HER-RAG framework with all functional modules enabled.

5. Experimental Results and Analysis

5.1 Overall quantitative results on TruthfulQA dataset

Table 1 summarizes overall experimental results obtained on the TruthfulQA benchmark.

Table 1. Experimental results on TruthfulQA dataset

Model	Factual-consistency F1	Hallucination AUC	Hallucination ratio (%)	ROUGE-L
Vanilla Llama-3-8B	0.619	0.643	47.2	0.547
Single-dense RAG	0.712	0.735	34.5	0.612
BM25-RAG	0.747	0.770	30.8	0.633
HER-RAG (Proposed)	**0.846**	**0.881**	**24.5**	**0.684**

The vanilla large-language-model baseline yields a high hallucination ratio reaching 47.2 %. Conventional RAG implementations achieve partial hallucination-suppression effects. The proposed HER-RAG reduces the hallucination-generation ratio down to 24.5 %. Compared with vanilla LLM and conventional single-dense-retrieval RAG baselines, the hallucination ratio decreases by 22.7 % and 13.5 %, respectively. The two-layer hybrid-retrieval component improves key-fact evidence recall capability; dynamic weighting and filtering eliminate noisy misleading snippets; factual-consistency auxiliary loss imposes generation-side constraints. Multiple modules cooperate to suppress hallucinated outputs jointly.

5.2 Quantitative results on HotpotQA multi-hop-reasoning dataset

Table 2 presents experimental results for multi-hop reasoning queries from HotpotQA. Multi-step reasoning raises overall task difficulty, and performance metrics degrade for all compared models. Nevertheless, HER-RAG still maintains optimal comprehensive performance. It demonstrates that two-layer hybrid retrieval can gather diversified evidence required for multi-step inference tasks, and dynamic weighting differentiates evidence importance for complex-query AIGC generation scenarios.

Table 2. Experimental results on HotpotQA dataset

Model	Factual-consistenc y F1	Hallucination ratio (%)	ROUGE-L
Vanilla Llama-3-8B	0.584	51.6	0.511
Single-dense RAG	0.676	39.3	0.576
BM25-RAG	0.703	35.7	0.594
HER-RAG (Proposed)	**0.817**	**27.1**	**0.652**

5.3 Ablation experimental results (TruthfulQA dataset)

Ablation studies are conducted to verify the individual contribution of each core module, as shown in Table 3.

Table 3. Ablation experimental results on TruthfulQA

Experimental	Factual-consistenc	Hallucination ratio	Explanation

Setup	y F1	(%)	
Single dense retrieval only	0.743	33.9	Insufficient recall of key factual-evidence materials
Two-layer retrieval, no dynamic weighting-filtering	0.775	31.2	Noisy evidence is retained and introduces misleading information
Two-layer retrieval + weighting, no factual-consistenc y loss	0.798	28.4	Prompt-only soft constraints cannot fully suppress hallucinations
Full HER-RAG framework	0.846	24.5	Optimal performance obtained via collaborative multi-module effects

Ablation-study outcomes validate that the three core components (two-layer hybrid-evidence retrieval, dynamic-evidence weighting-and-filtering, factual-consistency auxiliary loss) all deliver positive contributions for hallucination mitigation; none of them can be omitted without obvious performance degradation.

5.4 Long-context robustness test

We gradually increase the quantity of retrieved evidence fragments to simulate real-world long-context input scenarios. As the number of evidence snippets rises, the hallucination ratio of conventional RAG baselines increases rapidly. Benefiting from the dynamic-weighting filtering mechanism that discards large quantities of low-relevance fragments, HER-RAG exhibits a much slower growth rate of hallucination ratio and demonstrates better long-context robustness.

> **Figure 1 placeholder**: Long-context robustness comparison curve (x-axis: number of evidence snippets; y-axis: hallucination ratio). Insert actual plotted figure here.

6. Discussion

Limitations of the proposed framework:

1. The overall performance of HER-RAG heavily depends on the intrinsic quality of the external knowledge base. If the knowledge-base corpus itself contains erroneous factual entries, incorrectly retrieved evidence will still mislead AIGC generation outputs. Future research shall incorporate additional knowledge-base quality-verification sub-modules.

2. For completely novel facts that are not covered inside the external knowledge-base corpus, the model can only return the statement "No relevant materials available" and cannot execute open-ended creative reasoning for unknown facts.

3. Our main experiments adopt the open-source Llama-3-8B as the model backbone. When migrated to closed-source API-based large-language-model services, modifying model loss-function becomes unavailable; only retrieval-side and inference-phase weighting-filtering modules can be applied.

Future research directions:

1. Extend the framework to support multi-modal evidence inputs and realize hallucination-suppression for image-text mixed multi-modal AIGC generation.

2. Integrate LLM-Agent-style intelligent agents to implement dynamic incremental updates for external knowledge bases.

3. Construct domain-specialized knowledge bases targeting healthcare and legal vertical industries, to further improve factual reliability for high-risk domain-specific AIGC content-generation applications.

7. Conclusion

This paper presents HER-RAG, a hybrid retrieval-augmented generation framework designed to reduce factual inconsistency in AIGC text generation. The framework combines sparse BM25 and dense-vector retrieval, evidence scoring and filtering, weighted prompt construction, and a factual-consistency training objective. On the reported TruthfulQA and HotpotQA experiments, HER-RAG improves factual-consistency metrics and reduces the measured hallucination ratio relative to the evaluated vanilla Llama 3 8B and single-retriever RAG baselines. These findings

support the potential value of jointly improving evidence retrieval and generation-side factual constraints; however, reproducibility depends on fully specifying the evidence-scoring module, factual-consistency loss, retrieval configuration, training procedure, and hallucination-evaluation protocol.

References

AUTHOR ACTION REQUIRED: Verify every bibliographic entry against the original source, especially recent RAG/hybrid-retrieval and Llama-related references, and expand the review to include recent work on hybrid retrieval, reranking/evidence filtering, corrective/adaptive RAG, attribution/factuality evaluation, and hallucination mitigation. Format references according to the target journal.

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